



EMPLOYEE DATA ANALYSIS

Google Data Analytics Certificate Project

This project presents a comprehensive analysis of employee data using the six-phase Google Data Analytics methodology (Ask, Prepare, Process, Analyze, Share, Act).

Four main datasets were integrated: employee records, training data, engagement surveys, and recruitment data. A critical data quality issue was discovered and corrected in the employee status column, where 38% of records were misclassified. This revealed the true employee turnover rate to be 51.1%, not 12.9% as previously indicated.

The analysis identified training outcome as the strongest predictor of employee departure, with a 54.3% turnover rate among those who failed training compared to 48.6% among those who successfully completed it.

Python was used for data cleaning and analysis, with results exported to Tableau for interactive dashboard visualization. The report provides actionable recommendations to improve employee retention, focusing on training programs and high-risk departments.

Part One: Introduction

Project Title

Employee Turnover Analysis and Influencing Factors

Project Objective

This project aims to analyze employee data to understand the causes of employee turnover and identify factors influencing employee retention or departure, using the six-phase data analysis methodology (Ask, Prepare, Process, Analyze, Share, Act).

Data Sources

The dataset for this project was obtained from Kaggle (Employee/HR Dataset - All in One). It is a synthetic dataset designed for data analysis practice. The raw data includes four separate files covering employee demographics, training records, engagement surveys, and recruitment applications.

Link: <https://www.kaggle.com/datasets/ravindrasinghrana/employeedataset>

The dataset consists of four main files:

- Employee Data - 3,000 employees, 27 columns
- Training Data - 3,000 records, 9 columns
- Employee Engagement Survey Data - 3,000 surveys, 5 columns
- Recruitment Data - 3,000 applicants, 20 columns

Part Two: Data Cleaning

Key Data Quality Discoveries

Employee Status Column Inconsistency

Original Status	Total	Correct	Incorrect	Error Rate
Active	2,458	1,467	991	40.3%
Future Start	69	0	69	100%
Leave of Absence	86	0	86	100%
Terminated for Cause	66	66	0	0%
Voluntarily Terminated	321	321	0	0%

Conclusion: 38% of data (1,146 employees) was misclassified.

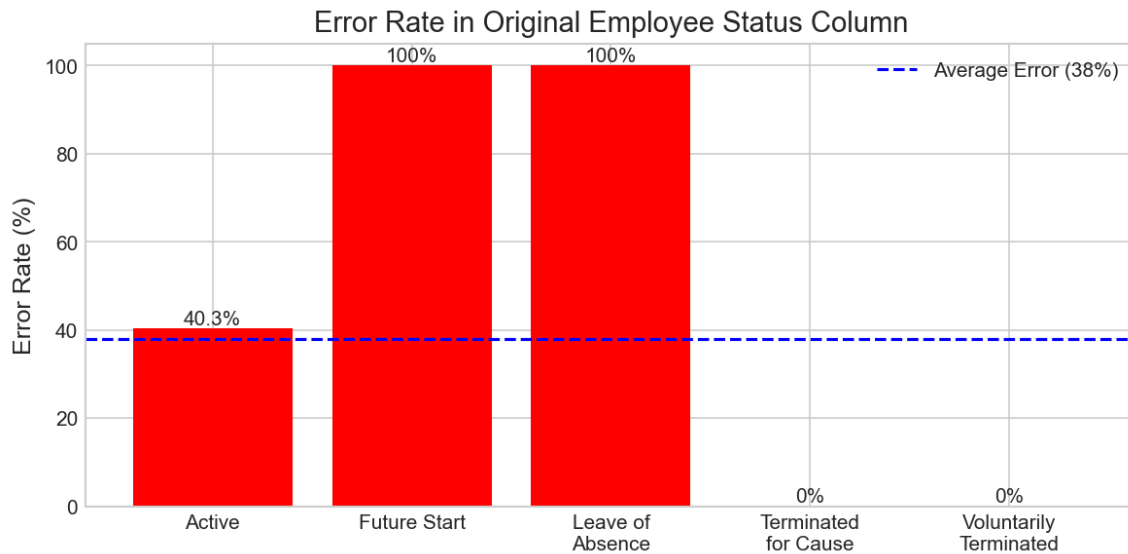


Figure 1: Error Rate in Original Employee Status Column

Impact on Turnover Calculation

- Using original column: 12.9% only
- After correction: 51.1%
- Difference: 38.2% (the true rate was underestimated by approximately 75%)

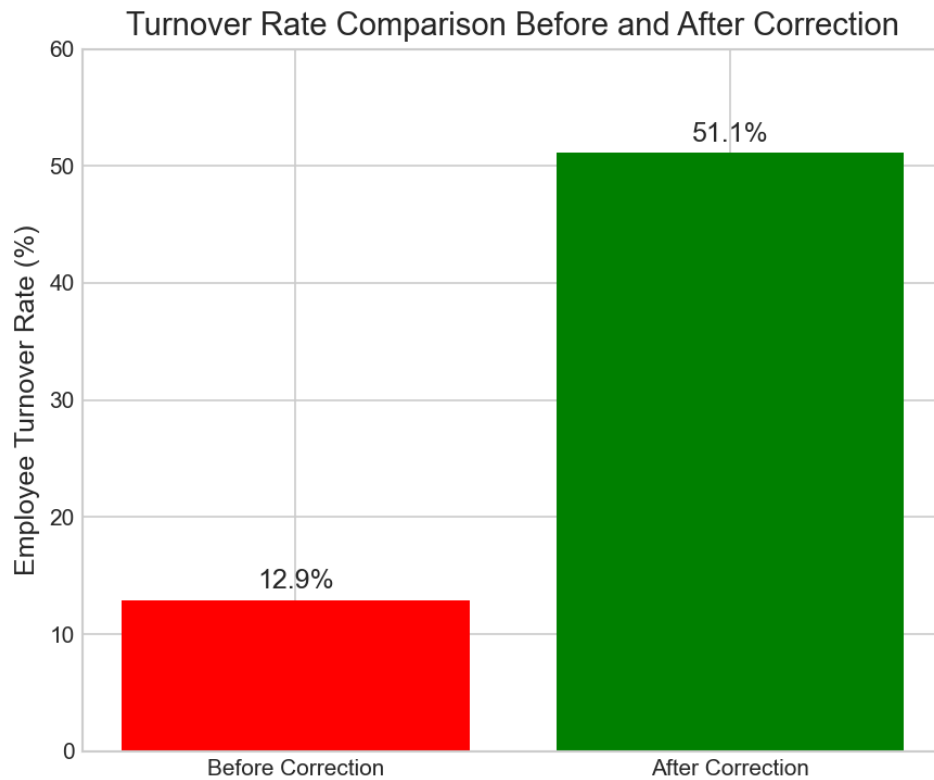


Figure 2: Turnover Rate Comparison Before and After Correction

Part Three: Exploratory Analysis

General Statistics

Metric	Value
Total Employees	3,000
Active Employees	1,467 (48.9%)
Terminated Employees	1,533 (51.1%)
Average Age	51.8 years
Average Performance Rating	2.97 out of 5
Average Satisfaction	3.02 out of 5
Average Engagement	2.94 out of 5

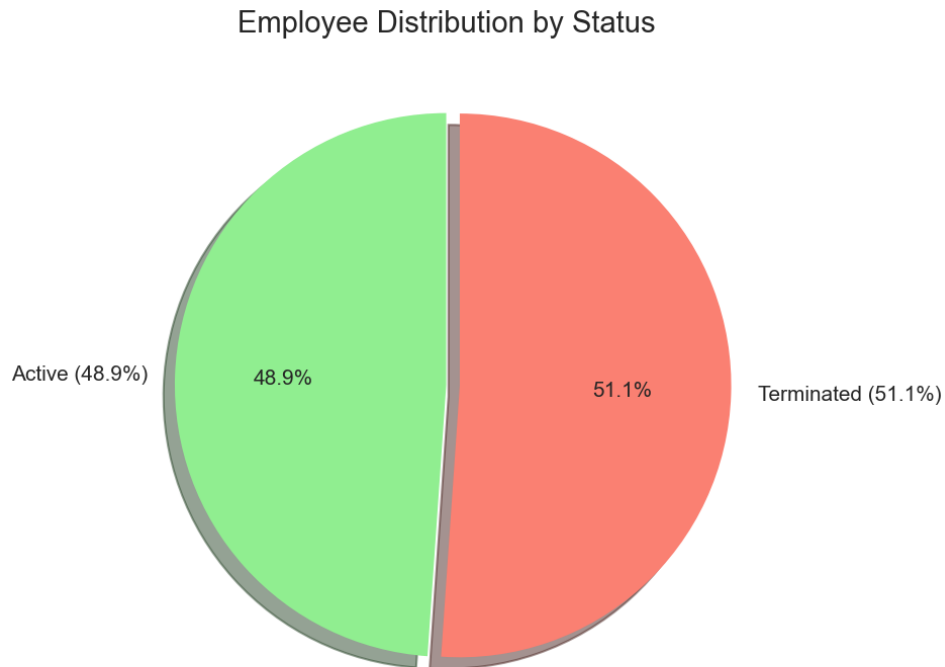


Figure 3: Employee Distribution by Status

Part Four: Key Findings

Employee Turnover by Department

Department	Active %	Terminated %	Status
Software Engineering	45.9%	54.1%	Highest
IT/IS	46.9%	53.1%	High
Admin Offices	47.0%	53.0%	High
Executive Office	47.1%	52.9%	High
Production	49.7%	50.3%	Medium
Sales	50.0%	50.0%	Medium

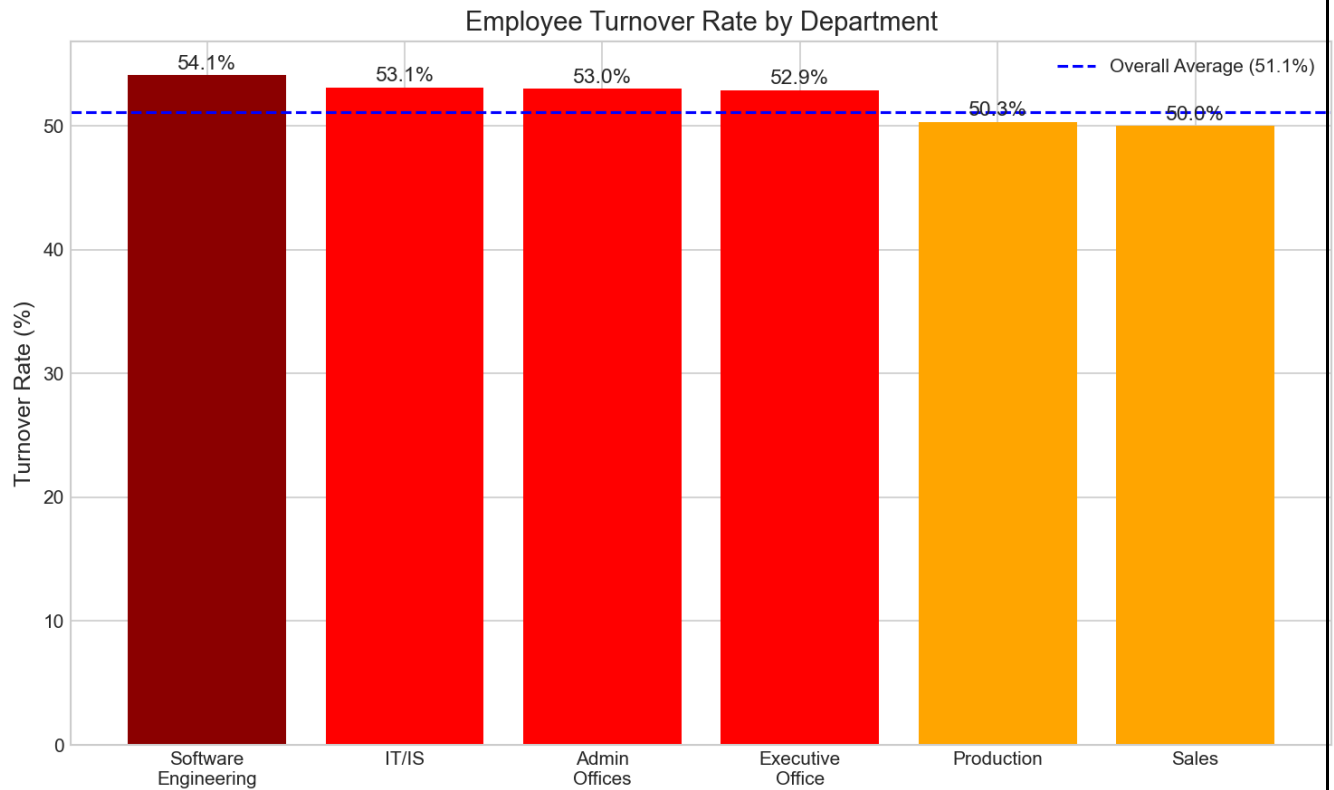


Figure 4: Employee Turnover Rate by Department

Termination Reasons by Department

Department	Voluntary	Retirement	Involuntary
Software Engineering	60.0%	20.0%	20.0%
Sales	53.3%	36.7%	10.0%
Executive Office	54.5%	18.2%	27.3%
Admin Offices	53.4%	20.6%	26.0%
Production	49.4%	25.5%	25.0%
IT/IS	49.1%	23.2%	27.7%

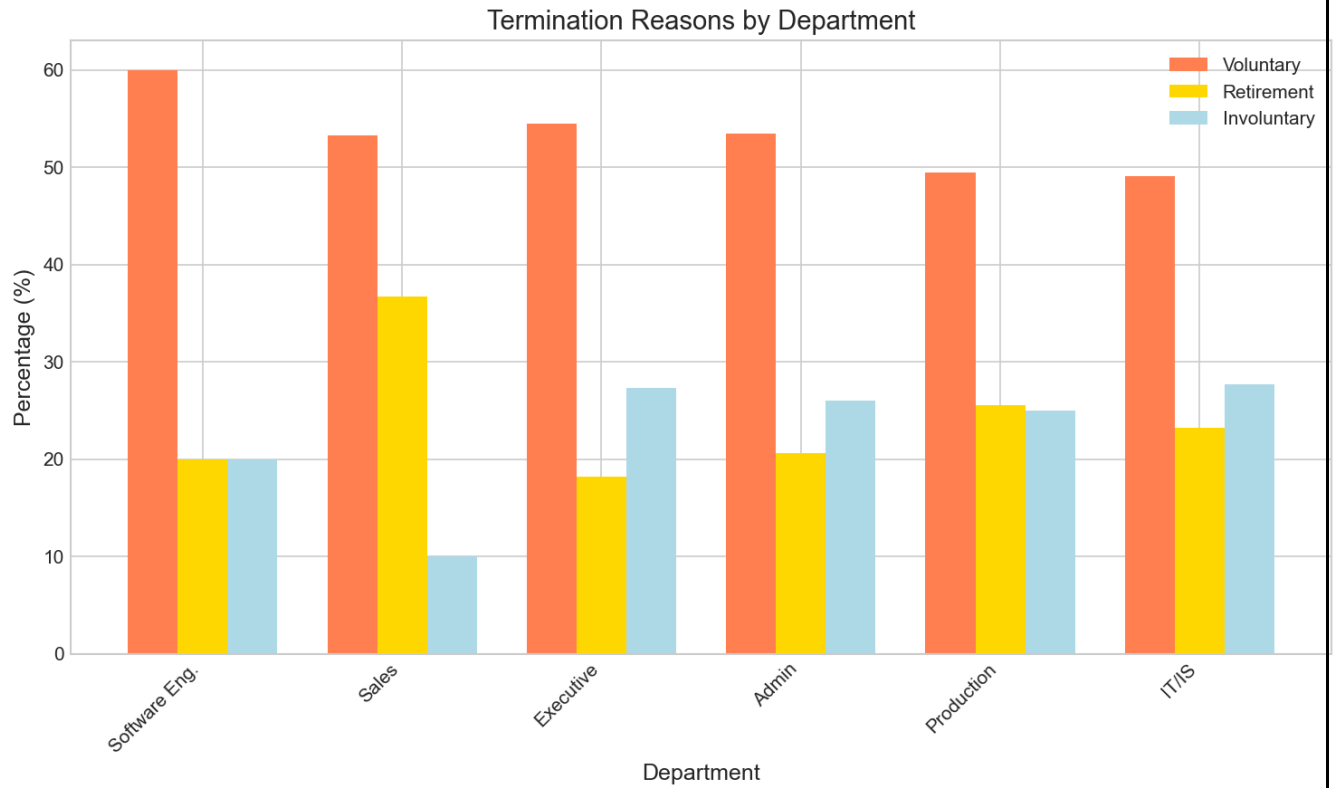


Figure 5: Termination Reasons by Department

Training Impact on Turnover

Training Outcome	Turnover Rate	Difference from Average
Failed	54.3%	+3.2%
Incomplete	52.9%	+1.8%
Completed	48.6%	-2.5%

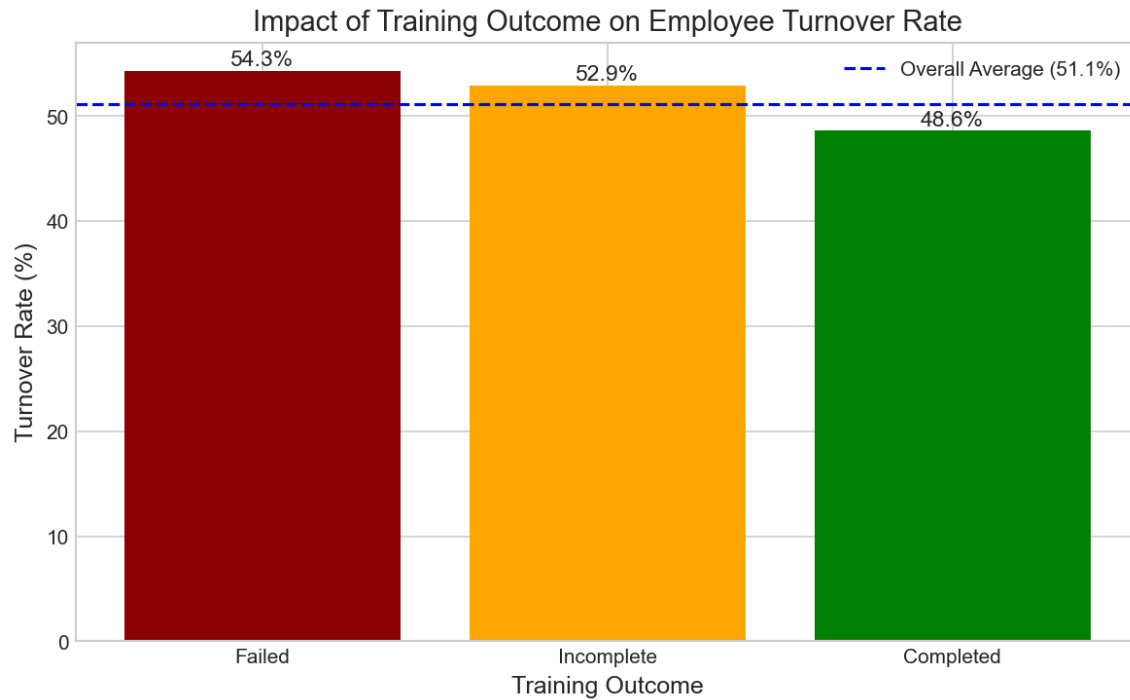


Figure 6: Impact of Training Outcome on Employee Turnover Rate

Part Five: Recommendations

Recommendation 1: Improve Training Programs

- Reason: Employees who fail training have 5.7% higher turnover rate
- Action: Redesign training programs in high-risk departments (Admin Offices, IT/IS)
- Goal: Reduce training failure rate from 25% to below 15%

Recommendation 2: Focus on Software Engineering Retention

- Reason: 60% of departures in this department are voluntary
- Action: Conduct exit interviews, improve work environment, review compensation
- Goal: Reduce voluntary turnover to 40%

Recommendation 3: Succession Planning for Sales Department

- Reason: 36.7% of departures are due to retirement
- Action: Implement knowledge transfer program and train new employees
- Goal: Ensure business continuity after retirement waves

Recommendation 4: Conduct Systematic Exit Interviews

- Reason: Current data doesn't explain "why" employees leave
- Action: Add qualitative questions to understand real reasons
- Goal: Gain deeper insights into departure drivers

Recommendation 5: Improve Data Quality

- Reason: 38% of employee status data was found to be incorrect
- Action: Automate periodic data validation processes
- Goal: Ensure accuracy and reliability of future data

Part Six: Tableau Dashboard

The cleaned data was exported to Tableau for interactive visualization. Four dashboards were created:

Dashboard 1: Turnover Overview

- Pie chart: Active (48.9%) vs Terminated (51.1%)
- Bar chart: Turnover rate by department
- Heat map: Turnover by department and gender

Dashboard 2: Training Impact

- Bar chart: Turnover by training outcome (Failed: 54.3%, Completed: 48.6%)

- Scatter plot: Training cost vs turnover rate

Dashboard 3: Department Deep Dive

- Dual axis: Satisfaction score vs turnover rate
- Stacked bar: Termination reasons by department

Dashboard 4: Recruitment Funnel

- Funnel chart: Application status flow
- Bar chart: Monthly applications (Peak: July)

Key Insights from Tableau:

The visualizations confirm that training outcome and department are the strongest predictors of turnover, while gender and age show no significant impact.

Part Seven: Conclusion

Project Summary

This project successfully analyzed employee data across four different files (employee, training, survey, recruitment).

A major data quality issue was discovered and corrected in the employee status column (38% errors), which changed the turnover rate from 12.9% to 51.1%.

Key Findings

1. Training outcome is the strongest predictor of turnover likelihood
2. Significant differences exist between departments in termination reasons
3. Age, gender, and performance are not influencing factors in this data

Study Limitations

The data used is synthetic data and may not reflect the actual reality of real organizations.

Correlations between variables were near zero, confirming the synthetic nature of the data.

Lessons Learned

- Importance of validating data quality before beginning analysis
- Necessity of understanding data context and logical relationships
- Value of documenting all cleaning and analysis steps

Next Steps for the Organization

Based on this analysis, the organization should:

1. Prioritize training program improvement in Admin Offices and IT/IS
2. Implement retention strategies for Software Engineering department
3. Begin succession planning for Sales department
4. Establish systematic exit interviews to capture qualitative data
5. Automate data quality checks to prevent future misclassification errors